

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/2024 Semester 1 MH1100 Calculus I

Tutorial 9, Week 10

Your tutor will aim to discuss: Problem 1, 2, 3, 4, 5, 6, 7, and 8

Problem 1 Verify the given linear approximation at $a = 0$.

(a)

$$\sqrt[4]{1+2x} \approx 1 + \frac{1}{2}x$$

(b)

$$\tan x \approx x$$

[Solution:] (a) Note that,

$$f'(x) = \frac{1}{4}(1+2x)^{-\frac{3}{4}} \cdot 2 = \frac{1}{2}(1+2x)^{-\frac{3}{4}}$$

We have $f(0) = 1$ and $f'(0) = 1/2$. The linear approximation function is

$$L(x) = f(a) + f'(a)(x-a) = f(0) + f'(0)x = 1 + \frac{1}{2}x$$

thus we have

$$\sqrt[4]{1+2x} \approx 1 + \frac{1}{2}x$$

(b) Notice that

$$f'(x) = \sec^2 x$$

We have $f(0) = 0$ and $f'(0) = 1$. The linear approximation function is

$$L(x) = f(a) + f'(a)(x-a) = f(0) + f'(0)x = 0 + 1 \cdot x = x$$

thus we have

$$\tan x \approx x$$

Problem 2 Use the linear approximation to estimate the given number

(a)

$$(1.999)^4$$

(b)

$$\sin 88^\circ$$

[Solution:] (a) We let $f(x) = x^4$ and let $a = 2$. The derivative is $f'(x) = 4x^3$ and $f(a = 2) = 16$, $f'(a = 2) = 32$. The function can be approximated by

$$f(x) \approx L(x) = f(a) + f'(a)(x-a) = 16 + 32(x-2)$$

Therefore

$$f(1.999) \approx 16 + 32 \cdot (1.999 - 2) = 15.968$$

(b) We let $f(x) = \sin x$ and let $a = \frac{\pi}{2}$. The derivative is $f'(x) = \cos x$ and $f(a = \frac{\pi}{2}) = 1$, $f'(a = \frac{\pi}{2}) = 0$. Notice that $88^\circ = \frac{88}{180}\pi = \frac{22}{45}\pi$. The function can be approximated by

$$f(x) \approx L(x) = f(a) + f'(a)(x - a) = 1 + 0(x - \frac{\pi}{2}) = 1$$

Therefore

$$f(88^\circ) \approx 1$$

Problem 3 When the blood flows along a blood vessel, the flux F (the volume of blood per unit time that flows past a given point) is proportional to the fourth power of the radius R of the blood vessel:

$$F = kR^4$$

Show that the relative change in F is about four times the relative change in R . (Note: here k is a constant.)

[Solution:] We denote the change in F as ΔF . It can be approximated by the differential

$$\Delta F \approx dF = 4kR^3 dR$$

The relative change in F is

$$\frac{\Delta F}{F} \approx \frac{dF}{F} = \frac{4kR^3 dR}{kR^4} = 4 \frac{dR}{R}$$

Thus the relative change in F is about four times the relative change in R .

Problem 4 Find the critical number of the function

(a)

$$f(x) = x^3 + x^2 - x$$

(b)

$$g(x) = |3x - 3|$$

[Solution:] (a) Note that the derivative always exists and equals to

$$f'(x) = 3x^2 + 2x - 1 = (3x - 1)(x + 1)$$

When $x = -1$ and $x = 1/3$, the derivative equals to zero. So they are two critical numbers.

(b) Notice that the derivative does not exist at $x = 1$, so it is a critical number.

Problem 5 Prove that the function

$$f(x) = x^{101} + x^{51} + x + 1$$

has neither a local maximum nor a local minimum.

[Solution:] Note that $f(x)$ is continuous and differentiable in the whole domain. Therefore, if $f(x)$ has a local extreme value at, say $x = c$, then $f'(c) = 0$. However, we have

$$f'(x) = 101x^{100} + 51x^{50} + 1 = 101(x^{50})^2 + 51(x^{25})^2 + 1 > 0.$$

So $f'(x)$ is always positive (nonzero) in the whole domain. In this way, there is no local extreme values.

Problem 6 Find the absolute maximum and absolute minimum of the function

$$f(x) = x - 2 \cos x$$

on the domain $[-2, 0]$.

[Solution:] We have

$$f'(x) = 1 + 2 \sin x$$

When $f'(x) = 0$, we have $x = -\pi/6 + 2\pi * N$ or $x = 7\pi/6 + 2\pi * N$. We find that only $x = -\pi/6$ is within the domain $[-2, 0]$. We evaluate the function values at critical number and boundaries,

$$f(-2) = -2 - 2 \cos(-2) \approx -1.168$$

$$f(0) = 0 - 2 \cos(0) = -2$$

$$f(-\pi/6) = -\pi/6 - 2 \cos(-\pi/6) = -\pi/6 - \sqrt{3} \approx -2.256$$

So $f(-\pi/6) = -\pi/6 - \sqrt{3}$ is absolute minimum, it obtained at $-\pi/6$. $f(-2) = -2 - 2 \cos(-2)$ is absolute maximum, it obtained at -2 .

Problem 7 Let $f(x) = (x - 3)^{-2}$. Show that there is no value of c in $(1, 4)$ such that $f(4) - f(1) = f'(c)(4 - 1)$. Why does this not contradict the Mean value theorem?

[Solution:] Note that

$$f'(x) = -2(x - 3)^{-3}$$

is defined in the whole domain except at $x = 3$, where the function and derivative do not exist.

From $f(4) - f(1) = f'(c)(4 - 1)$, we have $f'(c) = \frac{1 - 1/4}{3} = 1/4$. From the derivative equation, we have $-2(c - 3)^{-3} = 1/4$, however, $c = 1$ is not in the domain $(1, 4)$.

This does contradict with the mean value theorem, because function and the derivative is not defined at $x = 3$. So $f(x)$ is not always continuous and differentiable in the domain $(1, 4)$.

Problem 8 Use the mean value theorem to prove the inequality

$$\sin a - \sin b \leq |a - b|$$

for all a and b .

[Solution:] Since $f(x) = \sin(x)$ is continuous and differentiable in the whole domain, for any number a and b , and $a \neq b$, we have

$$\sin a - \sin b = \cos(c)(a - b)$$

with number c between a and b . So we have,

$$\sin a - \sin b = \cos(c)(a - b) \leq |\cos(c)(a - b)| = |\cos(c)||a - b| \leq |a - b|$$

Further, if $a = b$, the equation still holds.

Problem 9 Let $f(x) = 1/|x|$ and

$$g(x) = \begin{cases} 1/x & \text{if } x > 0 \\ 1 - 1/x & \text{if } x < 0 \end{cases}$$

Show that $f'(x) = g'(x)$ for all x in their domains. Can we conclude that $f - g$ is constant as in the Corollary of Mean Value Theorem?

[Solution:] Notice that

$$f(x) = \begin{cases} 1/x & \text{if } x > 0 \\ -1/x & \text{if } x < 0 \end{cases}$$

The derivative for both $f(x)$ and $g(x)$ is

$$f'(x) = g'(x) = \begin{cases} -1/x^2 & \text{if } x > 0 \\ 1/x^2 & \text{if } x < 0 \end{cases}$$

However, we should notice that both functions have no derivative at $x = 0$. Therefore we can have $f - g$ is constant in any domain which does not have $x = 0$ in it. If $x = 0$ is inside the region, we do not have $f' = g'$, since they do not even exist at $x = 0$.

Problem 10 Suppose that f is differential in the whole domain ($\mathbb{R}$) and has two roots, show that f' has at least one root?

[Solution:] If we denote the two roots of $f(x)$ as $x = a$ and $x = b$ and $a < b$, so we have $f(a) = 0$ and $f(b) = 0$. Since f is differential in the whole domain ($\mathbb{R}$), f is continuous on $[a, b]$ and differentiable on (a, b) . Further we have $f(a) = f(b)$, so from the Mean value theorem we have

$$f(b) - f(a) = f'(c)(b - a)$$

with c a value in the domain (a, b) . We can see from the above equation $f'(c) = 0$. That means f' has at least one root.

Extra Problem (Taylor polynomial) The tangent line approximation $L(x)$ is the best first-degree (linear) approximation to $f(x)$ near $x = a$, because $f(x)$ and $L(x)$ have the same rate of change (derivative) at a . For a better approximation than a linear one, we can consider a parabola (represented as $p(x)$) instead of a straight line. To make sure that the approximation is a better one, we requires the following conditions:

(1)

$$p(a) = f(a)$$

(2)

$$p'(a) = f'(a)$$

(3)

$$p''(a) = f''(a)$$

The parabola function $p(x)$ can be expressed in the form

$$p(x) = A + B(x - a) + C(x - a)^2$$

Show that the parabola function $p(x)$ that satisfies condition (1-3) is

$$p(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$$

From the expression of $p(x)$, we have $p'(x)$ expression

$$p'(x) = B + 2C(x - a)$$

and $p''(x)$ expression

$$p''(x) = 2C$$

Use condition (1) and $p(x)$ expression, we have

$$f(a) = p(a) = A$$

Use condition (2) and $p'(x)$ expression, we have

$$f'(a) = p'(a) = B$$

Use condition (3) and $p''(x)$ expression, we have

$$f''(a) = p''(a) = 2C$$

Therefore we have

$$p(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$$